sensors

MDPI

Article

Quality Control of Carbon Look Components via Surface Defect Classification with Deep Neural Networks

Andrea Silenzi [1], Vincenzo Castorani [2], Selene Tomassini [1], Nicola Falcionelli [1], Paolo Contardo [1], Andrea Bonci [1], Aldo Franco Dragoni [1] and Paolo Sernani [3,*]

[1] Dipartimento di Ingegneria dell'Informazione, Università Politecnica delle Marche, Via Brecce Bianche 12, 60131 Ancona, Italy; s1091549@studenti.univpm.it (A.S.); s.tomassini@pm.univpm.it (S.T.); n.falcionelli@pm.univpm.it (N.F.); p.contardo@pm.univpm.it (P.C.); a.bonci@univpm.it (A.B.); a.f.dragoni@univpm.it (A.F.D.)
[2] HP Composites Spa, Via del Lampo S.N., Z.Ind.le Campolungo, 63100 Ascoli Piceno, Italy; v.castorani@hpcomposites.it
[3] Department of Law, University of Macerata, Piaggia dell'Università 2, 62100 Macerata, Italy
* Correspondence: paolo.sernani@unimc.it

Abstract: Many "Industry 4.0" applications rely on data-driven methodologies such as Machine Learning and Deep Learning to enable automatic tasks and implement smart factories. Among these applications, the automatic quality control of manufacturing materials is of utmost importance to achieve precision and standardization in production. In this regard, most of the related literature focused on combining Deep Learning with Nondestructive Testing techniques, such as Infrared Thermography, requiring dedicated settings to detect and classify defects in composite materials. Instead, the research described in this paper aims at understanding whether deep neural networks and transfer learning can be applied to plain images to classify surface defects in carbon look components made with Carbon Fiber Reinforced Polymers used in the automotive sector. To this end, we collected a database of images from a real case study, with 400 images to test binary classification (defect vs. no defect) and 1500 for the multiclass classification (components with no defect vs. recoverable vs. non-recoverable). We developed and tested ten deep neural networks as classifiers, comparing ten different pre-trained CNNs as feature extractors. Specifically, we evaluated VGG16, VGG19, ResNet50 version 2, ResNet101 version 2, ResNet152 version 2, Inception version 3, MobileNet version 2 , NASNetMobile, DenseNet121, and Xception, all pre-trainined with ImageNet, combined with fully connected layers to act as classifiers. The best classifier, i.e., the network based on DenseNet121, achieved a 97% accuracy in classifying components with no defects, recoverable components, and non-recoverable components, demonstrating the viability of the proposed methodology to classify surface defects from images taken with a smartphone in varying conditions, without the need for dedicated settings. The collected images and the source code of the experiments are available in two public, open-access repositories, making the presented research fully reproducible.

Keywords: quality control; carbon fiber reinforced polymers; CFRP; deep learning; transfer learning; Industry 4.0

check for updates

Citation: Silenzi, A.; Castorani, V.; Tomassini, S.; Falcionelli, N.; Contardo, P.; Bonci, A.; Dragoni, A.F.; Sernani, P. Quality Control of Carbon Look Components via Surface Defect Classification with Deep Neural Networks. *Sensors* **2023**, *23*, 7607. https://doi.org/10.3390/s23177607

Academic Editors: Steve Vanlanduit and Roberto Teti

Received: 10 July 2023
Revised: 11 August 2023
Accepted: 30 August 2023
Published: 1 September 2023

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

In the fourth industrial revolution, commonly known as "Industry 4.0", data-driven methodologies such as Machine Learning (ML) and Deep Learning (DL) are recognized essential for the implementation of production systems capable of self-organizing, predicting (and correcting) their own faults, and adapting to variable human needs [1,2]. ML and DL are enabling several Industry 4.0 applications, such as predictive maintenance [3], anomaly detection [4], aided design [5], and many others [6].

A prominent example of an Industry 4.0 application is the automatic control of the quality of products and their components [7]. Indeed, defect detection technology is capable

中文详解 / 原文第1页

摘要：先理解论文回答的问题（精读）

[1] 作者关注汽车碳纤维外观件的表面质量：能否只用普通照片，把需要人工目检的外观质量判断自动化？这里不是检测飞机内部损伤，也不是用网络给出裂纹位置。输出是整张图片的类别。先把对象、输入、输出说清楚，再谈模型好不好。

[2] 摘要中的迁移学习，是借用ImageNet上训练好的视觉网络，再适配较小的工业数据。预训练不是预测：前者说参数来自哪里，后者说拿当前参数对新输入计算答案。作者比较十种CNN，属于实验比较，不是提出一个全新的基础网络。

[3] 摘要里接近97%的结果依赖特定数据、增强、模型和重复划分设置，不能直接拿它与我们的ResNet18单次83.75%做公平排名。我们要先对齐任务、样本、划分、增强和选模规则。

[4] 词语：CFRP是碳纤维增强聚合物；carbon look强调外观织纹。自检：用一句话说出这篇论文的输入和输出，能否输出缺陷轮廓？答案是这里的分类器不直接提供轮廓。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

of working in the long run with high precision, with obvious advantages over manual quality control [8]. In addition to the quality control of the final product, as in plastic injection molding [9,10], the detection of defects in the materials used in manufacturing is considered pivotal in many application domains, from the reliability of aircrafts to the efficacy of sanitary systems [11]. In this regard, the available literature focuses on integrating traditional Nondestructive [1] Testing (NDT) techniques, such as Infrared Thermography [12], with ML and DL methodologies, in order to detect structural and inner defects that might compromise the integrity of the composite materials.

Instead, this paper studies the use of deep Convolutional Neural Networks (CNNs) to classify surface defects in composite materials and specifically Carbon Fiber Reinforced Polymers (CFRP) for the automotive sector, using plain images of real defective parts, without integrating any NDT techniques. Given the difficulty of obtaining analytic models to automatically detect defects of CFRP-covered components, due to their complex structure, and the challenge of standardizing characterization and measuring of such defects [13], neural networks can help in addressing the challenge by being used in the quality control of such components [14]. The images used in the experiments presented in this paper come from the production of CFRP of "HP Composites Spa", a company located in Ascoli Piceno, Italy, and specialized in providing carbon fiber reinforced components. Specifically, in "HP Composites Spa", the quality control of carbon look components covered with CFRP is performed with the naked eye by specialized staff to check for surface defects such as weft discontinuities and porosities that might endanger the aesthetics of the components. Hence, we want to understand if deep CNNs are suitable to automate or integrate the quality control by automatically processing plain images of carbon fiber reinforced components. As we are interested in the appearance of carbon look components rather than the structural performance (which has been extensively covered in the literature, such as in pultrusion [15]), in this study, we aim at understanding whether a transfer learning [2] approach is adequate to classify the surface defects from smartphone pictures of the components. Therefore, we do not apply any NDT techniques requiring dedicated settings, usually implemented to detect structural or inner defects of components made of composite materials [16–19]. In this regard, transfer learning is suitable as it implies the use of models already trained on a large dataset to execute a different classification task, i.e., in a different feature space [20].

The research work performed on the "HP Composites Spa" case study and described in this paper adds the following contributions to the state of the art of surface defect detection in composite materials:

- A new database composed of plain images of CFRP-covered components for the automotive sector is introduced. The database is intended to train and benchmark techniques to classify defective and non-defective parts, as well as for multi-class classification into non-defective, recoverable, and non-recoverable components, using the pictures only, without thermography or any other NDT techniques. The database includes 400 images (200 per class) intended for the binary classification task and 1500 images (500 per class) for the multi-class classification. All the images are 224 × 224 pixels (96 ppi) in a JPEG format. The database is publicly released in an open-access GitHub repository (the image database is publicly available at: https://github.com/airtlab/surface-defect-classification-in-carbon-look-components-dataset, accessed on the 10th July 2023).
- A systematic comparison of ten models for the classification of surface defects is provided. The models are deep neural networks based on ten pre-trained Convolutional Neural Networks (CNNs), implemented to process the samples end to end, testing the effectiveness of transfer learning and fine tuning in the classification of the surface defects of carbon look components. Specifically, the tested pre-trained CNNs are VGG16 and VGG19 [21], ResNet50 version 2, ResNet101 version 2, and ResNet152 version 2 [22], Inception version 3 [23], MobileNet version 2 [24], NASNetMobile [25], DenseNet121 [26], and Xception [27]. The CNNs are combined with fully connected layers, trained from scratch on the proposed dataset. The source code

中文详解 / 原文第2页

引言：为什么普通照片值得研究（精读）

[1] 这一页延续工业检测背景，并把传统无损检测与外观检查区分开。无损检测包含红外、超声等成像，不损坏样品；普通RGB照片描述可见表面。相同“缺陷检测”四个字，可以指完全不同的传感数据和实际任务。

[2] 少量缺陷图片往往难以支撑大网络从随机参数训练。借用已有网络，是在数据少时利用先前学过的纹理、形状等表示。它并不保证跨到任何材料都有效；拍摄方式、缺陷大小和标签规则变化，仍然会造成问题。

[3] 读引言时追问作者的缺口：现有方法是否需要专用设备？数据是否真实、是否公开？本方法究竟降低哪一种使用门槛？把这些问题写成汇报的研究动机，不要只复述“深度学习发展很快”。

[4] 与自己项目的联系：老师给的高铁涂层照片和复合材料冲击超声不能先混成同一种任务。自检：为何表面照片分类的高准确率，不能证明能检测内部脱层？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

of the comparison is publicly available in a GitHub repository (the source code of the experiments is publicly available at: https://github.com/airtlab/surface-defect-classification-in-carbon-look-components-using-deep-neural-networks, accessed on the 10th July 2023).
- A real application of Industry 4.0 is demonstrated, proposing the use of DL to automate the control of surface defects of carbon look components.

The main innovation introduced by the proposed approach is the possibility of avoiding the dedicated settings to detect and classify surface defects as those required by the ML and DL techniques available in the related literature. In this regard, the significance of the proposed image database with respect to the literature relies in its use to train and test techniques to detect surface defects on plain images, without the use of NDT techniques. In addition, as it is based on CNNs that exploit transfer learning, the proposed approach has the potential to be integrated in handheld devices to run the classification with the trained models. In this way, the assessment of the aesthetics of the carbon look components would be automated and standardized, without requiring complex skills. Furthermore, to the best of our knowledge, this is the first work in surface defect classification for composite materials to publicly release the data and the source code of the tests, providing reproducible results. To this end, it is worth noting that while the weights of the pre-trained networks are available and linked in the source code repository, we do not provide the weights of the fine-tuned models.[1] In fact, our experiments are based on a stratified shuffle-split cross-validation strategy, where the networks are fine-tuned ten different times on diverse data, averaging the results got on the test sets. However, we do provide the code to run the cross-validation, making our experiments fully reproducible.

The rest of the paper is organized as follows. Section 2 presents a literature review about the use of deep learning for the detection of defects in composite materials, comparing our methodology to some related research works. Section 3 describes the database built for the experiments and the methodology followed to detect surface defects in the CFRP for the automotive sector. Section 4 presents the experimental evaluation of the surface defect detection applied to the CFRP used in HP Composites Spa, discussing the results. Finally, Section 5 draws the conclusions of this research.

2. Related Works[3]

Nondestructive Testing (NDT) includes a multitude of techniques for the evaluation of materials and components without causing damage [28]. Therefore, even the properties of composite materials are evaluated with different NDT techniques, such as Infrared Thermography, Ultrasonic Testing, Radiographic Testing, and many others [29].

Following the effectiveness demonstrated in many application fields, such as image processing, object detection, speech recognition, and pattern recognition in general [30], DL is being integrated with NDT techniques, especially for the detection of defects that might jeopardize the reliability of composite materials. To this end, therefore, we present the works which are the state of the art of the integration of DL and NDT techniques, and, as such, achieved the best performance. For example, Liu et al. [17,31] proposed to use DL to improve the existing Infrared Thermography techniques. Specifically, they use Generative Adversarial[2] Networks (GAN) as an image augmentation approach to enhance thermographic images in order to detect defects via Principal Component Analysis (PCA). They tested their methodology on specimens fabricated with intentionally formed defects. Differently, Bang et al. [16] suggested to use DL to directly detect defects by using Faster Region-based Convolutional Neural Network (Faster RCNN) on thermographic images. Specifically, they trained and validated an Inception V2 CNN [32] on 2802 thermographic images, obtained via data augmentation from 467 original images downloaded from the Internet. Then, they obtained a 75% average precision on 320 images of two composite specimens (carbon fiber fabric and randomly oriented glass fiber) with artificial defects. A similar study was conducted by Fang et al. [18] who used Mask-RCNN, an extension of Faster CNN, on a dataset composed of 500 images of two different materials, plexiglass

中文详解 / 原文第3页

贡献与相关工作：公开材料意味着什么（精读贡献，略读背景列表）

[1] 作者强调公开数据与代码，也明确没有提供每次微调后的最终权重。原因是实验进行了十次不同随机划分与训练，报告汇总结果。因此“公开代码”不等于下载就有一个已经训练好、可直接测试的最终模型。

[2] 相关工作提到GAN、PCA、Faster R-CNN等术语。GAN可生成或转换数据，PCA是一种降维分析，Faster R-CNN是检测框模型；它们在本页主要是交代别人的方法，不是你本周必须全部掌握的清单。

[3] 重点学会比较维度：对象材料、成像方式、真实还是人工缺陷、标签形式、评估设置、是否开放。不同论文的准确率若来自不同任务，不能只按数字大小排序。

[4] 汇报可以把贡献写成“真实工业外观数据、普通图像输入、预训练模型对比、开放实验材料”。这是作者的贡献表述，不意味着工业部署已经完成。自检：代码、数据和训练权重分别是什么？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

and CFRP. They achieved an 86.2% average accuracy in detecting defects. Wei et al. [19] developed a U-Net [33] variant to segment damages on around 8000 thermographic images, achieving the best F1 score of 92.74% in detecting the simulated defects on Carbon Fiber Reinforced Polymers (CFRP). Marani et al. [34] applied a CNN on seven videos (with various lengths) generated by Step Heating/Long Pulse Thermography on Glass Fiber Reinforced Polymers (GFRP). They achieved an average accuracy of 84.6% in classifying four defect classes. Of course, Infrared Thermography [1] is not the only NDT technique combined with DL. For example, Meng et al. [35] applied DL on ultrasonic signals, developing a CNN to classify defects on CFRP laminates. Their CNN-based classifier performed ultrasonic pattern recognition on 1000 samples produced on purpose (5000 were used for training) and was able to classify 10 defect classes with a 98% recognition rate. Gong et al. [36] applied a CNN to X-ray images to detect slag inclusions in aeronautics composite materials. However, due to the lack of defective components to include in the dataset, they trained their CNN on different data (using X-ray images of welded components) and then applied the trained model to the detection of slag inclusion defects in aeronautics composite materials. They achieved a 96.8% accuracy on 260 testing samples.

Despite the use of different techniques to detect defects in diverse materials, the described related works share some common traits:

- They apply DL on top of NDT techniques, such as Infrared Thermography, to detect structural and inner defects inside the composite materials.
- Most of the described works [16,17,31,35] record accuracy metrics in the detection of defect on specimens with artificial damages, produced on purpose.
- They do not publicly release the data and the source code of the experiments performed to collect the accuracy metrics.

The methodology proposed in this paper differs from the described works on these points. Specifically, we identify components that might compromise the aesthetics of the carbon fiber reinforced fabric by directly applying deep neural networks on images, without using any specific NDT technique. In fact, the suggested methodology circumvents the necessity for specific configurations needed by the NDT techniques. This characteristic distinguishes it from described related works. Combining DL with NDT techniques requires tailored settings to identify and categorize defects in CFRP components. These specialized configurations can be resource-intensive, requiring precise tuning of parameters and advanced expertise to set up properly. Our proposed approach simplifies this process, enhancing the accessibility and applicability of automated quality assessment in CFRP manufacturing. Therefore, in our work, we focus on surface defects in composite materials, following the results achieved, in general, in surface defect detection in other materials. In fact, DL achieved outstanding results in defect recognition [37] and has been successfully applied to the detection of surface defects in different materials, for example, on metal [38], steel [39,40], fabric [41,42], and wood [43,44]. Moreover, we evaluated our models on images of real defective parts [2] instead of introducing artificial defects. With respect to the works that use artificial defects to test their methodologies, in our work, we use images of real defects, trying to cope with the level of complexity and randomness that artificial defects might fail to replicate. They can vary greatly in shape, size, location, and context, unlike synthetic defects, which are usually standardized and less representative of the true range of possible flaws. By training and evaluating our models on real defective parts, we address the capacity to accurately recognize and classify an array of real-world defects, enhancing the robustness and practical utility of the classification models.

The proposed image database also reflects the listed differences with the related works. It contains real-world images of defective and non-defective components, with no synthetic data; in addition, it is composed of plain images only, i.e., it does not include images collected with thermal cameras or in other dedicated settings typical of NDT techniques. Moreover, we released the images included in the proposed database and the source code of experiments in two public open-access GitHub repositories, making our experiments fully reproducible.

中文详解 / 原文第4页

相关工作：表面外观与内部损伤的边界（略读，保留比较逻辑）

[1] 本页列举红外、超声、X射线与深度学习的组合。虽然都出现CNN，但输入有温度序列、声学信号、透射图像等，信号所包含的物理信息不同。表面RGB并不能自动提供内部损伤的真实标签。

[2] 作者认为真实生产缺陷比规则制造的人工缺陷有更多形状、位置和背景变化，这是对实验现实性的论证。但数据来自一个企业，仍可能有来源单一的限制；“真实”不等于覆盖全部工况。

[3] 现在不需要逐个复现这些相关工作。你的阅读任务是选出最接近自己数据的几篇，注明相近的是材料、任务还是成像方式。不要看到“航空”两个字就认定任务一致。

[4] 自检：为你的两批数据分别写出传感方式；如果一种没有明确，标待确认，而不是猜成RGB。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

3. Materials and Methods

The goal of this research is to understand whether we can classify surface defects in components made of composite materials (specifically those covered with CFRP) using deep neural networks on plain images without applying any NDT techniques with dedicated settings. In fact, the aesthetics of carbon look components are essential in the automotive sector, to the point that, in "HP Composites Spa", the quality of such components is manually checked with the naked eye by specialized operators.

With the aim of investigating whether deep neural networks can be successfully applied to images of surface defects taken with a smartphone, we built a database of plain images of carbon look components intended to benchmark classifiers of such surface defects. The database was split into two datasets to test both binary classification ("defect" vs. "no defect") and multi-class classification, distinguishing between defect types. To perform such classifications, we proposed an end-to-end classifier architecture using deep neural networks and transfer learning, comparing ten different pre-trained CNNs as feature extractors. To this end, Sections 3.1 and 3.2 describe the proposed database of images and deep neural network architecture.

3.1. Proposed Image Database

To evaluate the proposed end-to-end classifiers of surface defects, we built a database of plain images of the carbon look components produced in "HP Composite Spa". The images are 224[4] × 224 pixels (96 ppi) in the JPEG format. The images were originally taken at 3968 × 2976 pixels, then cropped at the center and resized at 224 × 224 to cope with the input size of the pre-trained CNNs used in this study. Moreover, the images depict components with carbon covered flat surfaces or with a slight curvature. The database is split into two datasets:

- One dataset is binary, including two classes. A total of 200 images are labeled as "no defect" (Figure 1a), as they are with no defects or present limited recoverable porosities[1]; 200 images are labeled as "defect" (Figure 1b), as they include weft discontinuities. This set of images is intended for binary classification, to test the performance of models that sort the components into defective or non-defective.
- The second dataset is multi-class, including three classes, i.e., "no defect", with recoverable defects, and with non-recoverable defects. The dataset contains 1500 images[2], 500 per class. The "no defect" class (Figure 2a) includes images of components without any surface defect. The recoverable defect class (Figure 2b) includes images of components with limited porosities and the infiltration of external materials (such as aluminum). Such defects can be treated and corrected. Finally, the non-recoverable defect class (Figure 2c) includes images of components with weft discontinuites, severe porosities, and resin accumulations. In "HP Composite Spa", these components are discarded, as their appearance cannot be recovered. With such dataset, we test the capability of the proposed models to classify multiple defect classes.

Specifically, the recoverable defects are:

- Isolated porosities, i.e., isolated holes in the surface of the material that only damage the aesthetic performance, but not the structural tightness (an example is provided in Figure 2b).
- Infiltration of foreign objects (aluminum or polyethylene) on the material surface that can be removed.

Instead, the non-recoverable defects are:

- Severe porosities, i.e., where the holes in the surface of the material are not isolated and cover most of the surface (an example is provided in Figure 2c).
- Weft discontinuities, i.e., all the cases in which the characteristic texture of the interwoven carbon fiber bundles is altered, generally caused by a wrong overlapping of the materials or poor fiber adhesion to the mold (an example is provided in Figure 1b).

中文详解 / 原文第5页

数据定义：二分类和三分类不是同一标准（必须精读）

[1] 二分类共有400张，200张no defect与200张defect。关键是no defect不仅包含完全没有缺陷的图片，也包含有限、允许修复的孔隙；defect主要对应织纹不连续。因此negative/positive是业务标签，不是"消极/积极"的因素，也不是数学中图片数值的正负。

[2] 三分类共有1500张，每类500张：无缺陷、可修复、不可修复。孤立孔隙或可移除异物可进入可修复类；严重孔隙、织纹断续、树脂堆积等进入不可修复类。这里的可修复主要依企业对外观恢复的判断，不代表航空结构强度的修理认证。

[3] 类别标签，就是给图片附一个标准答案。程序把名称映射为整数，例如negative=0、positive=1；这个数字没有大小优劣含义。分类网络学习整张图属于哪一类，不从这个整数获得缺陷的位置。

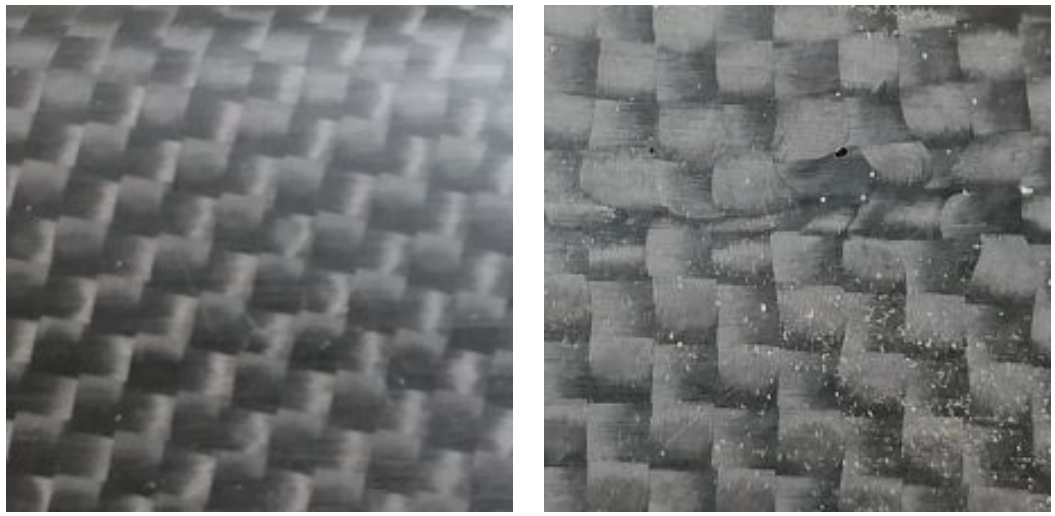
[4] 图像最初较大，之后中心裁剪并缩放到224×224。尺寸越小可能丢掉细小缺陷，裁剪也可能漏掉边缘；预处理不是随便改变外观，而是需要兼顾网络输入与缺陷保留。

[5] 与你的程序对应：data.py按子目录读取类别，每类200张；model.py定义名称顺序。我们检查发现二、三分类之间有重复图，不能拿一个训练后在另一个上声称完全独立泛化。自检：为什么二分类negative不能等价于三分类无缺陷?

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

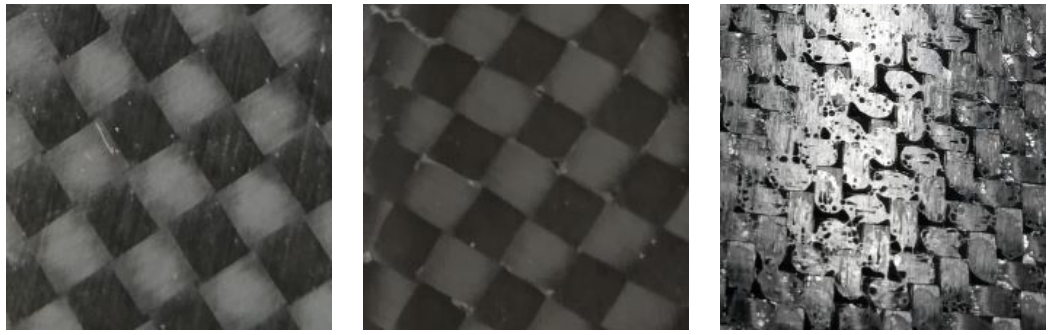
- Accumulations of resin caused by the imperfect calibration of the spaces between the two fiber molds and the silicone mandrel interposed between them.

The images are real-world examples of all the surface defects that "HP Composites Spa", the company that provided the images, need to classify.



(**a**) (**b**)

Figure 1. Negative (**a**) and positive (**b**) samples from the binary dataset. The negative sample includes no defect. Instead, the positive sample has a weft discontinuity at the center and porosities in the carbon fiber reinforced fabric covering the component.



(**a**) (**b**) (**c**)

Figure 2.[3] Negative (**a**), recoverable defect (**b**), and non-recoverable defect (**c**) class samples from the multi-class dataset. The negative samples has no defects. The recoverable defect sample includes spare porosities. The non-recoverable defect sample includes severe porosities and weft discontinuities.

The images were manually labeled[1] according to the recommendations of the domain experts, i.e., the HP Composite staff, in particular to distinguish between recoverable and non-recoverable defects. The images were taken from different angles[2] and distances, without standard lighting. In this way, we emulated the conditions that an operator, in charge of controlling the appearance of the components, would face using a handheld device for the quality control, without dedicated settings as in NDT. Thus, we investigated the capability of the proposed models of working under varying conditions in order to assess the viability of our approach based on plain images taken in a non-standard, non-fixed environment.

We trained and tested the models proposed in this paper on both datasets, in their original form. Moreover, we ran the classification tests on an augmented version of the datasets. In fact, despite the fact that 400 images and 1500 images are available to test the classification with two classes and three classes, respectively, deep neural networks usually

中文详解 / 原文第6页

图1、图2：看懂样例及标注来源（必须精读）

[1] 图1展示二分类的负类和正类，图2展示三分类样本。先看织纹连续性、孔隙稀疏程度和杂质；再核对图注，不要仅靠肉眼重新给标签。数据标签来自企业领域人员对质量标准的判断，是训练时参考的答案。

[2] 论文说图片来自不同距离、角度与非标准照明，试图接近手持目检。对网络来说，阴影、反光、视角也会改变像素；它可能学到这些关联，而不完全是缺陷。因此后续错误分析要检查误报是否集中在反光区域，漏检是否集中在小缺陷。

[3] 本页没有为每张图提供框或像素轮廓。整图类别只能训练分类任务；要研究缺陷定位或面积，通常还需要新增框或掩膜，不能把模型的注意可视化直接当成真实标注。

[4] 你看图时记录四列：文件名、真实类别、你观察到的现象、疑问。不确定样例不要擅改标签，而是形成待问导师或领域人员的问题。自检：同一类别为什么可以有不同外观？不同拍摄角度能否算不同独立工件？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

require a higher number of training images to reduce the classification error. To face such issue, data augmentation has been proven useful to cope with the scarcity of image data and overfitting problems [45]. To this end, in our experiments, we augmented the datasets by flipping[5] the original images on both axes. Thus, we built an augmented binary dataset composed of 1200 images (600 per class): 400 are the original images, 400 are the same images, but flipped on the horizontal axis, and 400 are flipped on the vertical axis. Similarly, we built an augmented multi-class dataset composed of 4500 images (1500 per class). We applied flipping (i.e., a 180° rotation on both axis) without using other augmentation techniques (such as rotation and scaling) to check the capability of the proposed model to converge to a proper training with a relatively small set of data for fine-tuning. In addition, adding other augmentation techniques to the collected dataset could have caused overfitting in the models, as it would have increased the replication of the same data.

3.2. Proposed Classification Model

Figure 3 shows the architecture of the end-to-end models proposed in this paper. To classify the plain images of carbon look components, we compared ten models based on ten deep CNNs pre-trained on the Imagenet database [46] used as a feature extractor, combined with a classifier based on fully connected dense layers. In fact, we applied a transfer learning methodology, i.e., the use of models already trained on a large dataset to execute a different classification task. In addition to dealing with the scarcity of training data, transfer learning can achieve better generalization than a dedicated training from scratch and prevent overfitting [47,48]. In this regard, transfer learning based on the pre-training on ImageNet has been recognized as particularly effective for general-purpose classifiers [49]. While the database proposed in this paper is too small to train the CNNs from scratch, training from scratch on images of surface defects of carbon look components would be feasible, at the risk of introducing overfitting. Therefore, we compared ten different[3] pre-trained CNNs, namely VGG16, VGG19, ResNet50V2, ResNet101V2, ResNet152V2, InceptionV3, MobileNetV2, NASNetMobile, DenseNet121, and Xception. We removed from the pre-trained CNNs the fully connected layers, averaging the output of the last convolutional layers with Global Averaging Pooling[1]. The obtained feature vector[2] was used as the input for fully connected dense layers composed of neurons with the Rectified Linear Unit (ReLU) activation function. The final classification was performed by a dense layer using the Softmax[4] activation function, composed of two neurons for the classification on the binary dataset, and three neurons for the multi-class dataset.

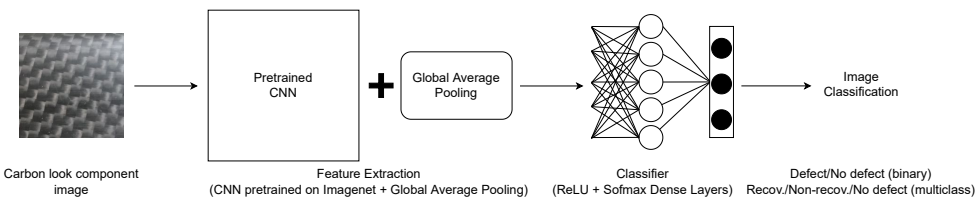


Figure 3. The architecture proposed to classify the plain images of carbon look components. The input is processed by a CNN pre-trained on the Imagenet database. The CNN, without its final fully connected layers, is combined with a Global Average Pooling layer and used to extract features. Then, fully connected layers are added and trained from scratch on the dataset. The final classification is performed by the last dense layer, with the Softmax activation function. Different combinations of fully connected layers (and hyperparameters) have been tested to improve the final classification (see Tables 1 and 2).

We trained each model in an end-to-end fashion on the proposed datasets, fine-tuning the pre-trained CNNs and training the fully connected layers from scratch. Specifically, we looked for the best hyperparameters in terms of classification accuracy on the binary dataset and the multi-class dataset. We tuned the following parameters:

中文详解 / 原文第7页

图3：特征提取器、全局池化和分类器（必须精读）

[1] 图3把输入图片送入预训练CNN，卷积层输出一组特征图，再经全局平均池化得到特征向量，送入新分类层。特征图可以理解成很多响应图，每个通道表示网络学到的一种响应模式；它不是人预先手写的“孔隙分数”。

[2] 全局平均池化把每个通道在空间位置上的响应取平均。例如最后有512个通道，每个通道平均成一个数，就得到512维向量。空间位置被汇总，适合整图分类，但不能据此直接得到缺陷轮廓。

[3] 论文比较VGG、ResNet V2、MobileNet、DenseNet、Xception等十种骨干。注意：我们的ResNet18并不是文中列出的模型，且只用一个新Linear层；论文还比较不同隐藏层，因此目前是入门基线，不是完整复现。

[4] 分类器为每类产生分数，Softmax把分数变成和为1的概率形式。输出2个数对应二分类，输出3个数对应三分类。它们不是两个独立的“有缺陷程度”，而是互相竞争的类别输出。

[5] 本页同时谈翻转增强：变换增加观察形式，不增加独立试件。你自己的新实验应先划分原图，再仅对训练组增强，并按工件来源分组检查，防止同一原图的翻转版本出现在测试组。本文是否充分保证这一点，需要看实现，不能仅靠描述下结论。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

- The number of the CNN final layers to be fine-tuned on the proposed dataset, testing 8, 4, and 0.[1] This means that, during the end-to-end training of the model, the weights were freezed for the starting layers of the pre-trained CNNs, while for the last 8 (or 4 or none) layers, the weights were modified with the backpropagation.
- The optimizer to perform the error backpropagation, testing the Stochastic Gradient Descent (SGD) with a 0.9 momentum and Adam. For both, we compared different learning rates,[2] i.e., 0.001 and 0.0001.
- The use of Batch Normalization[4] for regularization between the Global Average Pooling and the first dense layer.
- The number of fully connected layers to be added to the pre-trained CNNs to perform the final classification. Specifically, we tested a single dense layer composed of 512 ReLU neurons followed by a final layer with Softmax activation, and two dense layers composed of 256 and 128 ReLU layers, followed by the Softmax layer.

To test the impact of the hyperparameters on the classification performance, we used the same evaluation protocol described in Section 4.1, based on a stratified shuffle split cross-validation repeated ten times. For all the models, we used the same batch size, i.e., 32 images, and the same number of training epochs, i.e., 100, applying an early stopping[3] strategy after 5 epochs without lowering the validation loss to avoid the overfitting.

To this end, Table 1 lists the configurations of hyperparameters which offered the best results in terms of classification accuracy among those tested on the binary dataset, including the fully connected layers used for classification and trained from scratch on the proposed dataset. In most of the models, the last 8 layers of the pre-trained CNNs were fine-tuned. Instead, the model based on MobileNetV2 offered its best results by freezing the weights of all the layers in the pre-trained CNN, training only the dense layers added for the final classification. With NASNetMobile and Xception, only the last 4 layers were fine-tuned. The models based on VGG16, VGG19, and InceptionV3 offered the best classification accuracy using the SGD optimizer for training. Instead, the other seven models obtained the best performance with Adam. The Batch Normalization did not add any improvement on the classification accuracy. Finally, the models based on MobileNetV2, NASNetMobile, and Xception offered the best results using two consecutive dense layers with 256 and 128 ReLU neurons. The other seven models obtained the best metrics with the dense layer composed of 512 ReLU neurons.

Table 1. The hyperparameters and the final layers of each model which obtained the best results on the binary augmented dataset.

	Fine-Tuning	Optimizer	Learning Rate	Batch Normalization	Dense 512 Units	Dense 256 Units	Dense 128 Units
VGG16	Last 8	SGD	0.0001	-	✓	-	-
VGG19	Last 8	SGD	0.001	-	✓	-	-
ResNet50V2	Last 8	Adam	0.0001	-	✓	-	-
ResNet101V2	Last 8	Adam	0.001	-	✓	-	-
ResNet152V2	Last 8	Adam	0.0001	-	✓	-	-
InceptionV3	Last 8	SGD	0.001	-	✓	-	-
MobileNetV2	0	Adam	0.001	-	-	✓	✓
NASNet	Last 4	Adam	0.0001	-	-	✓	✓
DenseNet121	Last 8	Adam	0.0001	-	✓	-	-
Xception	Last 4	Adam	0.0001	-	-	✓	✓

Table 2 includes the combinations of hyperparameters with the best results in terms of classification accuracy on the multi-class dataset. All the models achieved the top accuracy

中文详解 / 原文第8页

表1：训练设置与可训练参数（必须精读）

[1] 作者比较最后8层、4层或0层微调。冻结表示不更新这些旧权重；0层微调并不表示完全不训练，因为新加的分类器仍然需要学习。微调层数不是卷积次数，也不是训练轮数。

[2] 表1同时涉及优化器、学习率、BatchNorm、分类层宽度。SGD与Adam是更新参数的方法；学习率是更新步长；512或256/128是隐藏层神经元数量，不是图片数量。不能把超参数与卷积核中的可训练权重混为一谈。

[3] 论文的最大轮数100、批大小32，验证损失连续5轮不改善就提前停止；我们是最多30轮、批大小16、耐心7轮，且冻结整个骨干。因此这些差异都必须在结果对照中列明，不能简单声称差距只因训练轮数。

[4] BatchNorm还有运行均值、方差等状态。只设置requires_grad=False不一定冻结它们；我们的engine.py让整个骨干保持eval，只将新分类层按训练/评估切换。这解释了代码为什么不是简单调用model.train()。

[5] 自检：模型参数、超参数和运行统计各举一个例子。哪些由梯度更新？哪些由你在config.json中指定？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

by fine-tuning the last 8 layers of the pre-trained CNNs and using a single dense layer with 512 ReLu neurons. The Batch Normalization improved the results for the models based on VGG16 and VGG19. These two models and the one based on MobileNetV2 used the SGD optmizer for the training, whereas the other seven models obtained the best metrics with Adam.

Table 2. The hyperparameters and the final layers of each model which obtained the best results on multi-class augmented dataset.

	Fine-Tuning	Optimizer	Learning Rate	Batch Normalization	Dense 512 Units	Dense 256 Units	Dense 128 Units
VGG16	Last 8	SGD	0.001	✓	✓	-	-
VGG19	Last 8	SGD	0.001	✓	✓	-	-
ResNet50V2	Last 8	Adam	0.001	-	✓	-	-
ResNet101V2	Last 8	Adam	0.001	-	✓	-	-
ResNet152V2	Last 8	Adam	0.0001	-	✓	-	-
InceptionV3	Last 8	Adam	0.0001	-	✓	-	-
MobileNetV2	Last 8	SGD	0.0001	-	✓	-	-
NASNet	Last 8	Adam	0.0001	-	✓	-	-
DenseNet121	Last 8	Adam	0.0001	-	✓	-	-
Xception	Last 8	Adam	0.0001	-	✓	-	-

4. Experimental Evaluation

We compared the ten proposed end-to-end deep neural networks on the collected image database, evaluating accuracy metrics for the binary and the multi-class classification. In this section, we report the details about the results achieved by the best configuration (i.e., the set of hyperpameters, among those tested, which achieved the best accuracy) of each classifier, as explained in Section 3.2. We aim at understanding whether the pre-trained CNN-based networks can be used to detect and classify the surface defects of carbon look components, in the HP Composite case study and in general. Indeed, in addition to the creation of a baseline of metrics on the collected images, a comparison of classifiers might be relevant to other case studies about surface defect detection and classification in composite materials.

Therefore, in the following subsections, we describe the experimental setup as well as the metrics (Section 4.1) evaluated to compare the classifiers. Moreover, we present and discuss the results of the evaluation (Section 4.2). Finally, we analyze the limitations of our experimental evaluation (Section 4.3).

4.1. Evaluation Protocol and Metrics

We tested the proposed end-to-end deep neural networks on the binary and multi-class datasets, using both the original version and the augmented one, by applying a stratified shuffle split cross-validation scheme. In this regard, we repeated a randomized 80-20 split ten times, using the 80% of the data as the training set, and the 20% as the test set, preserving the percentage of samples from each class, in each split. A total of 12.5% of the training images, i.e., 10% of the whole dataset, was used as validation data in each split. To implement a fair comparison, the splits were the same for all the tested models. In this way, we tested the generalization capability of our models, making sure that they did not overfit on a specific data split. Table 3 includes the number of training, validation, and test images for each split of the described cross-validation scheme.

中文详解 / 原文第9页

表2与评估协议：公平比较首先靠数据划分（必须精读）

[1] 表2给出三分类的较优设置，不能照搬二分类的最优组合。这说明任务与标签改变后，最合适的训练方式也可能改变。先固定基线和评估方法，再讨论调参，不把全表所有组合同时展开。

[2] 论文重复十次分层随机划分。每次先留20%测试，再从剩余80%中取12.5%作为验证，即整体70%训练、10%验证、20%测试。分层表示保留各类别比例；不同模型使用同样的划分，避免某个模型恰好拿到容易的测试组。

[3] 我们的单次400张实验对应280/40/80。十次划分得到的是十次评估结果，不是把同一个模型连续训练十遍，也不意味着不同测试组完全互斥。均值和标准差反映这些设置下的表现与波动。

[4] 分层随机划分不自动解决同工件相关性。若同一部件不同视角进入训练和测试，仍可能高估跨工件能力。你后续需要尽量取得试件编号，并确定研究真正希望泛化到什么新对象。

[5] 自检：从80%的训练候选中取12.5%，为什么验证集占总数据的10%？算式是0.8×0.125=0.1。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

Table 3. Number of training, validation, and test images for each dataset in each randomized split of the stratified shuffle split cross-validation scheme.

Dataset	Total	Training	Validation	Test
Original Binary	400	280	40	80
Augmented Binary	1200	840	120	240
Original Multi-class	1500	1050	150	300
Augmented Multi-class	4500	3150	450	900

As highlighted in the Introduction section, a Jupyter notebook with the described experiments is available in a public GitHub repository in order to guarantee the reproducibility of the tests. The experiments ran on a virtual machine in a cloud environment, equipped with an 8-core Intel Xeon CPU E5-2623 v4 (2.60 GHz) (Intel Corporation, Santa Clara, CA, USA), 30 GB of RAM, and a Nvidia Quadro M4000 GPU (Nvidia Corporation, Santa Clara, CA, USA), using Keras 2.6.0, TensorFlow 2.6.0, and scikit-learn 1.0.2.

We compared the proposed end-to-end networks by measuring the average testing accuracy over the ten splits of the cross-validation scheme. Therefore, in each iteration, we computed the ratio between the samples which are correctly classified and the total number of samples in the test set. Moreover, we aggregated the results in each split to compute:

- The precision for each class, i.e., the ratio between the number of samples correctly classified as belonging to a class and the total number of samples labeled as that class in the test set.
- The recall for each class, i.e., the ratio between the number of samples correctly classified as belonging to a class and the total number of samples available for that class in the test set.

Precision and recall can be formulated in terms of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) according to the following equations:

$$precision = \frac{TP}{TP + FP}, \quad (1)$$

$$recall = \frac{TP}{TP + FN}. \quad (2)$$

In the binary classification, to compute the precision and recall of a class, the samples of that class are considered positive, whereas the samples of the other class are considered negative. In the multi-class classification, precision and recall for a C class generalize to the following equations:

$$precision_C = \frac{T_C}{T_C + F_C}, \quad (3)$$

$$recall_C = \frac{T_C}{T_C + F_{\not C}}, \quad (4)$$

where T_C is the number of samples correctly classified as C, F_C represents the number of samples labeled as C but actually belonging to a different class, and $F_{\not C}$ is the number of samples actually belonging to C but labeled with a different class.

Finally, we report the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) for the best model on both dataset to provide a complete picture of the diagnostic capabilities of the end-to-end networks. Considering the "no defect" as the negative class and the "defect" as the positive class, the ROC curve shows the true positive rate (TPR, i.e., the recall of the "defect" class) against the false positive rate (FPR, i.e., 1 minus the recall of the "no defect" class) when the classification threshold varies, for all the splits of the cross-validation. In the multi-class classification, the ROC curve is computed micro-averaging the ROC curves obtained for each label by binarizing the output (i.e., by considering one label as positive and all the other as negative).

中文详解 / 原文第10页

表3、公式1—4：精确理解评价指标（必须精读）

[1] 表3把四种数据版本的数量列清楚。原始二分类400张分为280/40/80；增强后二分类1200张分为840/120/240。增加的图片包括同图翻转，不能当成新增独立工件。

[2] 以positive为关注类：TP是确实为positive且预测为positive，FP是negative却报成positive，FN是positive却被漏掉。Precision=TP/(TP+FP)，问“报出的有多少是真的”；Recall=TP/(TP+FN)，问“该找的找回多少”。

[3] 用你30轮结果：TP=36、FP=9、FN=4、TN=31。查准率36/45=80%，召回率36/40=90%，准确率(36+31)/80=83.75%。这三个数字回答不同问题；不能把准确率直接叫召回率。

[4] 三分类可以轮流把每一类当成关注类，其余类合并为另一类，再算各类指标。宏平均让每类同等重要，微平均先合并计数；类别比例不平衡时两者可能相差很大。

[5] ROC曲线通过改变阈值比较真阳性率与假阳性率，AUC是曲线下的面积，不等于固定阈值下的准确率。论文使用不同划分的曲线展示波动；我们目前四张结果图没有ROC，不能说已经实现全部论文指标。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

4.2. Results and Discussion

In this subsection, we analyze the metrics obtained by the proposed neural networks on the collected image database, discussing the results on the binary classification task ("no defect" vs. "defect") and the multi-class classification task ("no defect" vs. "recoverable defect" vs. "non-recoverable defect"). We also report on the impact of the data augmentation on the accuracy of the proposed neural networks.

4.2.1. Results on the Binary Classification Task

Table 4 includes the class precision and recall obtained by the end-to-end networks based on different pre-trained CNNs on the original binary dataset. The metrics are averaged across the ten splits of the stratified shuffle split cross-validation scheme. With 140 training images per class, the end-to-end networks struggle to converge to a good classification result for both classes, given that precision and recall are below 90% for all the models. In fact, the results vary across different data splits, with ResNet101V2 obtaining the highest standard deviation [1] in the recall of the "no defect" class (17.64%). For example, in one of the splits, the network based on ResNet101V2 wrongly classifies 24 "no defect" samples out of 40 as containing defects, achieving a recall of 35%, whereas all the 40 "defect" samples are correctly classified, with a 100% recall. This means that, on such data split, the network tends to label most of the samples as defective. On the contrary, in a different data split, the ResNet101V2-based model correctly classifies 39 out of 40 "no defect" samples (recall 97.5%), but wrongly labels 30 out of 40 "defect" samples (recall 25%), tending to label the samples as "no defect". The best classifier is the network based on Xception [2], achieving the highest recall (87.64%) for the "no defect" class, and the highest precision (87%) and recall (89.75%) for the "defect" class. Xception also obtains the second-best precision (89.57%) for the "no defect" class, with the lowest standard deviation (2.69%). This means that the Xception-based model is capable of correctly classifying around 90% of the testing samples, independently from the label. The second-best model is the one based on DenseNet121, obtaining scores which are two percentage points below those of Xception. The other models achieve lower precision and recall, with higher standard deviation.

Table 4. Average class precision and recall, with standard deviation, on the original binary dataset. The metrics are computed over the 10 splits of the stratified shuffle split cross-validation scheme. The best values are highlighted in bold.

	No Defect		Defect	
	Precision	**Recall**	**Precision**	**Recall**
VGG16	85.26 ± 4.08%	79.00 ± 9.23%	81.03 ± 6.48%	86.00 ± 5.15%
VGG19	85.01 ± 6.89%	81.25 ± 8.16%	82.71 ± 6.58%	84.50 ± 9.60%
ResNet50V2	87.68 ± 5.60%	77.75 ± 7.37%	80.26 ± 5.84%	89.00 ± 4.77%
ResNet101V2	**89.89 ± 12.43%**	60.00 ± 17.64%	71.18 ± 8.56%	88.25 ± 21.51%
ResNet152V2	82.36 ± 4.79%	78.25 ± 6.23%	79.54 ± 4.13%	82.75 ± 6.37%
InceptionV3	83.86 ± 5.87%	73.25 ± 10.31%	76.80 ± 6.94%	85.50 ± 6.60%
MobileNetV2	83.83 ± 4.85%	75.75 ± 5.13%	77.96 ± 3.26%	85.00 ± 5.81%
NASNet	84.97 ± 5.64%	76.25 ± 7.00%	78.62 ± 5.32%	86.25 ± 5.94%
DenseNet121	87.22 ± 4.62%	85.25 ± 3.61%	85.60 ± 3.04%	87.25 ± 5.41%
Xception	89.57 ± 2.69%	**87.64 ± 4.70%**	**87.00 ± 5.68%**	**89.75 ± 3.05%**

Therefore, the results on the original binary dataset do not support the idea of classifying plain images of carbon look components into two classes, i.e., with defect and with no defect, at least when few training samples are available. With 140 training images per class, transfer learning is not effective for such binary classification task.

中文详解 / 原文第11页

表4：原始二分类结果为什么不稳定（精读结论，选两行算指标）

[1] 表4报告十次划分的每类查准率和召回率，形式为均值±标准差。均值概括整体，标准差提示对划分是否敏感；某次最好值不能代替十次平均。

[2] 文中Xception在原始二分类上整体较好，但并非所有网络都稳定。一个模型可能召回正类很多，却把大量负类误报为正类，因此必须同时看两类指标。

[3] 对你来说，先挑一种模型和一类读懂一行，不必背全表。逐项问：这个值属于哪个类、哪种指标、哪个数据版本、是否重复评估？缺一项，数字的意义就容易读错。

[4] 不要把作者对该任务的可行性判断直接推广到高铁或飞机。自检：为什么标准差较大时，单次测试容易给人过分乐观的印象？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

However, a significant improvement in the collected metrics occurs with data augmentation. To this end, Table 5 lists the average values of class precision and recall computed on the augmented binary dataset. For example, the recall of the "no defect" class increases from 60% ± 17.6% to 94 ± 3.1% for the model based on ResNet101V2. In general, all the networks are much more stable, with standard deviations significantly lower than those obtained on the original dataset.

Table 5. Average class precision and recall, with standard deviation, on the augmented binary dataset. The metrics are computed over the 10 splits of the stratified shuffle split cross-validation scheme. The best values are highlighted in bold.

	No-Defect		Defect	
	Precision	**Recall**	**Precision**	**Recall**
VGG16	93.71 ± 2.10%	93.00 ± 3.01%	93.10 ± 2.82%	93.75 ± 2.12%
VGG19	94.45 ± 3.69%	93.42 ± 3.34%	93.58 ± 2.97%	94.33 ± 4.15%
ResNet50V2	94.92 ± 3.40%	94.75 ± 2.91%	94.86 ± 2.56%	94.75 ± 3.93%
ResNet101V2	92.58 ± 3.14%	94.00 ± 3.09%	94.00 ± 2.87%	92.33 ± 3.45%
ResNet152V2	92.43 ± 2.08%	91.67 ± 2.86%	91.82 ± 2.43%	92.42 ± 2.37%
InceptionV3	94.44 ± 4.12%	91.33 ± 3.21%	91.62 ± 3.01%	94.50 ± 4.38%
MobileNetV2	**96.88 ± 1.17%**	95.42 ± 1.50%	95.50 ± 1.43%	**96.92 ± 1.18%**
NASNet	94.36 ± 2.42%	92.33 ± 2.20%	92.53 ± 1.98%	94.42 ± 2.58%
DenseNet121	96.82 ± 2.43%	**97.33 ± 1.66%**	**97.33 ± 1.63%**	96.75 ± 2.59%
Xception	96.63 ± 1.37%	97.17 ± 1.55%	97.18 ± 1.50%	96.58 ± 1.46%

The impact of data augmentation[2] is evident in Figure 4, which reports the average accuracy (and its standard deviation) of each model across the ten splits of the cross-validation, in the original and in the augmented binary datasets. The best model for the binary classification task on the augmented dataset is the one based on DenseNet121, achieving a 97.04%[1] (± 1.7) average accuracy, increasing by 11% its performance obtained on the original dataset. Xception scores a similar average accuracy (96.88%) but with a lower standard deviation (0.86%). MobileNetV2 also obtains a 96.17% accuracy (± 1.05%). The model based on ResNet101V2 is the one with the biggest improvement from the original dataset to the augmented one, improving from a 74.12% (± 6.4%) average accuracy to 93.17% (± 2.09%). In addition, the impact of data augmentation is highlighted by the ROC curve reported in Figure 5a for the original dataset and in Figure 5b for the augmented dataset, for the DenseNet121-based model. In the augmented dataset, the curve shows a lower variance, i.e., a greater stability of the classifier across the ten splits of the stratified shuffle split cross-validation. The average AUC is also greater (99.25 ± 0.56%) than that of the original dataset (94.08 ± 2.42%).

中文详解 / 原文第12页

表5：增强后结果与证据边界（必须精读）

[1] 增强后的二分类指标整体提高，DenseNet121平均准确率约97.04%，Xception约96.88%。这些是论文特定实验的报告值，不是保证你运行同数据就应达到的门槛。

[2] 文中把改善归因于数据增强，但可靠解释还要核对数据划分与增强的先后关系、相关样本是否跨组、调参是否独立于测试。增强不会凭空创造新的独立缺陷来源，这个检查是我们的评估建议，不是断言论文已发生泄漏。

[3] 与你的83.75%比较前，列出差别：骨干ResNet18与DenseNet121不同，分类头不同，是否微调不同，是否增强不同，重复次数不同，预处理也可能不同。只有控制了这些因素，才能判断改动造成什么影响。

[4] 一个合理下一步是预先指定训练组增强，保持验证和测试不变，比较两种设置；但你已看过当前测试结果，不能不停据此挑参数。正式研究要规划独立评估。自检：为什么翻转后的3倍图片不等于3倍新工件?

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

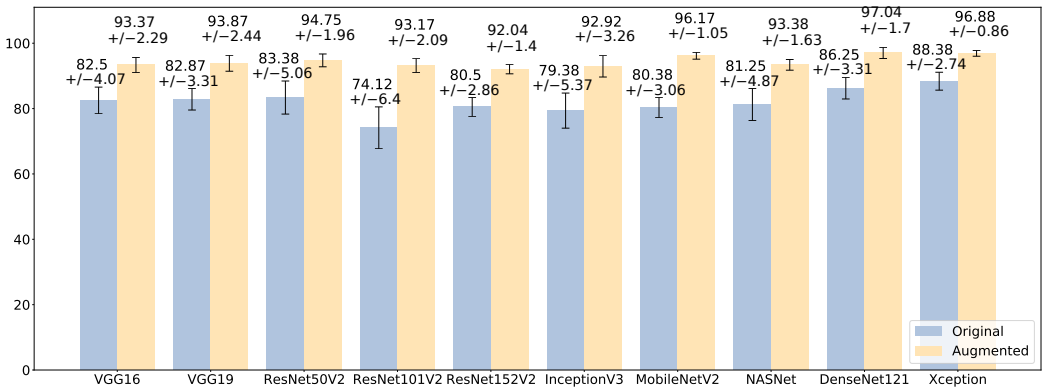
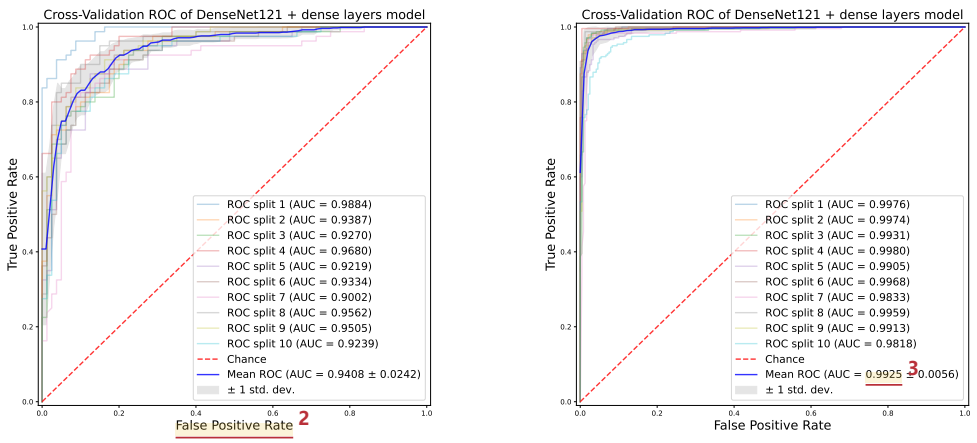


Figure 4.[1] Comparison of the average accuracy (%), with standard deviation, on the original binary dataset (blue) and the augmented one (yellow).

Therefore, the results achieved on the augmented dataset allow concluding that the classification of plain images into containing defects or not is feasible from smartphone images. The results are promising, as the best models (DenseNet121, Xception) achieve around 97% accuracy across all the splits of the cross-validation scheme.



(**a**) (**b**)

Figure 5. ROC curve for each split of the stratified shuffle split cross-validation scheme for the DenseNet121-based model, on the original binary dataset (**a**) and the augmented one (**b**).

4.2.2. Results on the Multi-Class Classification

The results on the multi-class classification exhibit a trend similar to those of the binary classification: the classification accuracy improves when using the augmented dataset instead of the original one. To this end, Table 6 shows the average class precision and recall on the original multi-class dataset, computed across the ten splits of the stratified shuffle split cross-validation scheme. All the ResNet-based models cannot converge to a proper classification on all the classes, interpreting as "recoverable" or "non-recoverable" images with no defect. Their classification capability strongly depends on the way data are randomly split. For example, ResNet50V2 exhibits a 21.20% standard deviation on the 61.60% recall for the "no defect" class. On the original dataset, the models which seem to converge to the classification of samples are those based on VGG16, VGG19, and DenseNet121 with the precision and recall between 86% and 93% for each class. Instead, all the other models obtain lower metrics.

中文详解 / 原文第13页

图4和图5：准确率柱图与ROC各回答什么（精读图，不背全部数值）

[1] 图4对比原始与增强数据的平均准确率，并给出标准差。看图先确定横轴模型、纵轴百分数、颜色含义和误差条，再判断差异是否稳定；不同模型误差条重叠时，不能只看柱高就宣称明显优越。

[2] 图5的横轴是假阳性率，纵轴是真阳性率。曲线越靠左上，在这些样本上越能兼顾较少误报和较多检出。多条曲线对应十次划分，不是十种模型；阴影提示划分间变化。

[3] 作者报告DenseNet121增强后平均AUC约0.9925。AUC不是99.25%的实际分类准确率，也没有自动决定现场可接受的阈值。现场还需结合漏检代价、误报处理成本和独立样本。

[4] 汇报图时用"图的目的—读图顺序—支持的结论—不能支持的结论"四句话。自检：一个AUC很高的模型，为什么在默认阈值下仍可能漏掉缺陷？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

Table 6.[1] Average class precision and recall, with standard deviation, on the multi-class original dataset. The metrics are computed over the 10 splits of the stratified shuffle split cross-validation scheme. The best values are highlighted in bold.

	No Defect		Recoverable		Non-Recoverable	
	Precision	**Recall**	**Precision**	**Recall**	**Precision**	**Recall**
VGG16	88.05 ± 2.52%	**90.80 ± 2.96%**	**89.93 ± 4.36%**	**91.10 ± 3.11%**	**93.00 ± 3.12%**	88.50 ± 4.50%
VGG19	**88.89 ± 3.39%**	89.40 ± 4.50%	89.70 ± 3.76%	89.40 ± 3.80%	91.95 ± 3.78%	91.30 ± 2.57%
ResNet50V2	82.61 ± 5.45%	61.60 ± 21.20%	76.31 ± 7.41%	79.70 ± 9.58%	76.09 ± 12.47%	88.50 ± 2.69%
ResNet101V2	79.19 ± 3.94%	74.80 ± 11.04%	80.93 ± 6.92%	77.80 ± 5.36%	81.38 ± 5.39%	87.60 ± 5.68%
ResNet152V2	74.03 ± 4.89%	72.60 ± 12.03%	83.20 ± 7.15%	69.90 ± 6.49%	72.96 ± 9.34%	84.00 ± 6.13%
InceptionV3	82.21 ± 3.83%	79.60 ± 2.73%	83.27 ± 1.90%	83.60 ± 4.76%	86.21 ± 2.72%	88.20 ± 2.44%
MobileNetV2	78.01 ± 3.18%	88.60 ± 3.01%	85.14 ± 3.03%	83.00 ± 3.19%	91.00 ± 2.78%	80.60 ± 5.10%
NASNet	78.76 ± 3.02%	77.50 ± 3.61%	84.52 ± 2.99%	83.70 ± 3.16%	83.04 ± 2.52%	84.70 ± 3.55%
DenseNet121[2]	86.02 ± 2.85%	87.10 ± 3.48%	89.90 ± 3.02%	88.40 ± 2.80%	91.37 ± 3.53%	**91.40 ± 2.94%**
Xception	83.92 ± 3.95%	87.10 ± 3.70%	87.57 ± 2.41%	87.30 ± 3.72%	90.46 ± 3.79%	86.90 ± 2.88%

The classification capability of the proposed models significantly improves when using the augmented dataset, as shown in Table 7. The models based on DenseNet121, VGG16, and VGG19 obtain more than 95% of precision and recall for all the classes. In general, all the models display better performance than the original dataset.

Table 7. Average class precision and recall, with standard deviation, on the multi-class augmented dataset. The metrics are computed over the 10 splits of the stratified shuffle split cross-validation scheme. The best values are highlighted in bold

	No Defect		Recoverable		Non-Recoverable	
	Precision	**Recall**	**Precision**	**Recall**	**Precision**	**Recall**
VGG16	95.54 ± 1.36%	**97.93 ± 0.94%**	**97.96 ± 1.00%**	**96.77 ± 1.21%**	96.80 ± 1.11%	95.50 ± 1.39%
VGG19	95.29 ± 1.51%	97.13 ± 1.14%	97.17 ± 0.88%	95.93 ± 1.39%	96.61 ± 0.88%	95.93 ± 1.67%
ResNet50V2	89.20 ± 3.61%	85.93 ± 6.42%	88.28 ± 5.15%	90.97 ± 2.48%	90.23 ± 4.49%	89.87 ± 6.23%
ResNet101V2	87.27 ± 3.72%	85.80 ± 9.26%	91.78 ± 4.46%	87.97 ± 3.84%	87.81 ± 4.97%	91.97 ± 3.63%
ResNet152V2	89.37 ± 3.36%	86.93 ± 4.78%	89.07 ± 2.96%	90.47 ± 2.81%	89.84 ± 4.84%	90.17 ± 4.32%
InceptionV3	93.51 ± 1.77%	95.13 ± 1.28%	95.78 ± 0.61%	94.60 ± 1.55%	94.96 ± 1.44%	94.43 ± 1.41%
MobileNetV2	91.05 ± 0.79%	94.63 ± 1.26%	95.45 ± 1.31%	92.70 ± 1.33%	94.21 ± 1.78%	93.17 ± 1.34%
NASNet	90.97 ± 1.20%	92.47 ± 2.14%	93.38 ± 1.57%	91.20 ± 0.56%	92.71 ± 1.33%	93.27 ± 1.21%
DenseNet121	**95.85 ± 1.38%**	97.20 ± 0.78%	97.81 ± 0.91%	96.30 ± 1.22%	**97.31 ± 0.96%**	**97.40 ± 1.10%**
Xception	94.08 ± 1.72%	96.77 ± 0.88%	97.32 ± 1.24%	95.83 ± 1.54%	96.33 ± 1.24%	94.97 ± 1.57%

The good performance with the augmented dataset, in comparison with the original one, is evident in Figure 6, which reports the average accuracy across the ten splits of the cross-validation scheme. The best model in the multi-class classification is the one based on DenseNet121, which has the best average accuracy, 96.97%, and the lowest standard deviation, 0.59%. The models based on VGG16 and VGG19 are very close, with 96.73 (±0.64) and 96.33% (±0.85) respectively. The model based on Xception, which was the second best in the binary classification, also exhibits a promising accuracy, scoring 95.86% (±0.74). Hence, in addition to very good accuracy, the low standard deviation highlights that these models are independent from the data splits of the cross-validation, demonstrating

中文详解 / 原文第14页

表6、表7：三分类与每类性能（精读比较结构）

[1] 原始三分类中，部分网络的无缺陷召回波动较大；增强后三类指标总体更好。表格每组列对应一个类别，不要把可修复类的查准率与不可修复类的召回率混在一起比较。

[2] DenseNet121、VGG16、VGG19在该增强任务中较强。模型更深并不一定在小数据上更好：优化难度、预训练匹配、参数规模和样本分布都可能影响结果，不能只按层数挑模型。

[3] 三分类仍然是企业外观质量标准。若你的高铁数据标签是划痕、气泡、针孔等检测框，类别语义和任务都变了，不能直接把三分类最后一层搬过去就认为完成迁移。

[4] 读表时挑无缺陷与不可修复两类，看哪种错误更重要，再把业务代价说清楚。自检：总准确率一样的两个三分类模型，为什么可能在实际质量控制上价值不同？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

their generalization capability. The good impact of data augmentation on the classifiers is depicted in Figure 7, which includes the ROC curves across the cross-validation splits of the model based on DenseNet121 on the multi-class dataset. The diagnostic capability of the classifier clearly increases in the augmented dataset, with the curves tending to the upper left corner of the diagram independently from a specific data split.

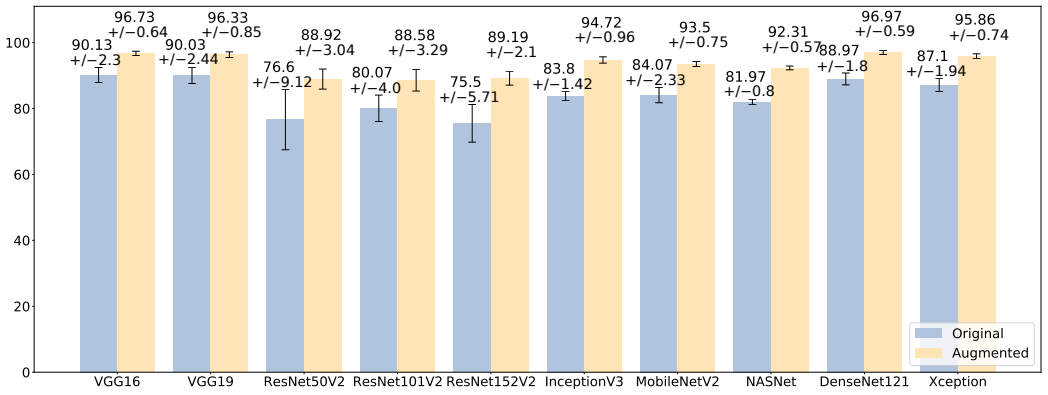
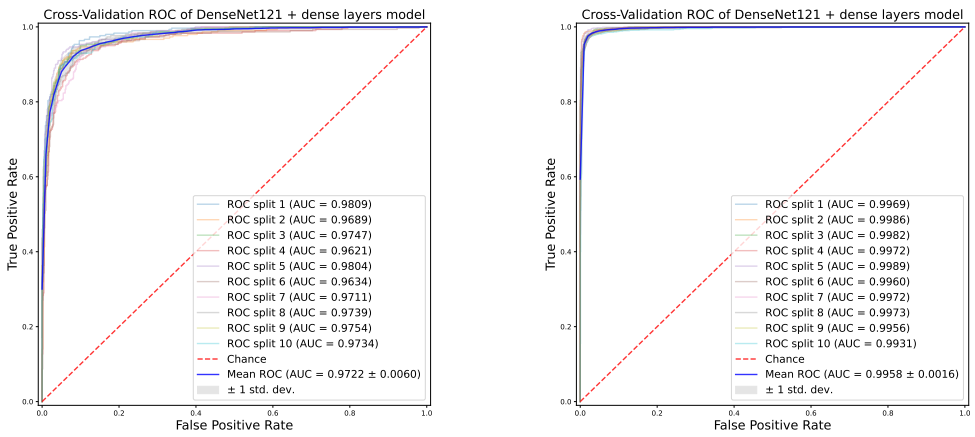


Figure 6. Comparison of the average accuracy (%), with standard deviation, on the original multi-class dataset (blue) and the augmented one (yellow).

These results allow concluding that classifying surface defects of carbon look components from plain images is definitely possible and transfer learning can be applied; different deep neural networks based on pre-trained CNNs are able to classify 900 images into recoverable, non-recoverable or those with no surface defects. The simplicity of the proposed models and the good accuracy achieved on images taken from different points of view, with varying lighting, suggest that handheld devices [2] could be used in practice, in addition to the naked-eye quality control of carbon look components currently performed in the company that provided the data for our experiments. In this way, a member of staff could spot the defect and use the proposed methodology to classify it in a standardized fashion. In fact, such architectures can be deployed in mobile devices or in the cloud and used to process smartphone pictures during quality control.



(**a**) (**b**)

Figure 7. ROC curve for each split of the stratified shuffle split cross-validation scheme for the DenseNet121-based model, on the original binary dataset (**a**) and the augmented one (**b**).

4.3. Limitations [3]

The results are promising both in terms of binary classification ("defect" vs. "no defect") and multi-class classification ("recoverable" vs. "non-recoverable" vs. "no defect"), but present some limitations. The proposed datasets include 400 images for the binary

中文详解 / 原文第15页

图6、图7与局限入口（精读图表及局限）

[1] 增强三分类的平均准确率中，DenseNet121约96.97%，VGG16约96.73%，VGG19约96.33%。这是特定任务的十次均值。数值相近时还要看波动、计算成本和误检类别，不能仅按小数点排名。

[2] 本页把分类可行性与手持部署联系起来，但真实上线还需要速度、设备性能和现场验证。图7为三分类ROC；图注出现与正文任务不一致的地方，应结合章节和数据说明核对，阅读时不盲从单行图注。

[3] 你的一次论文汇报可以在结果后立刻转到局限：单一企业、样本小、缺少独立工件分组、没有部署时延验证。把这些写成待验证问题，不能不经实验就当作自己的创新成果。

[4] 自检：论文中"可能用于手机端"与"已经在手机端实际部署并验证"有什么区别？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

classification task and 1500 images for the multi-class classification task. Data augmentation solves the scarcity of training images and is, in general, recognized useful to limit overfitting. However, testing with more original images[1] of diverse components would allow drawing more general conclusions.[4] This limitation can be addressed by collecting even more real-world images to expand the dataset and furtherly limit overfitting. Nevertheless, to the best of our knowledge, the database released with this paper is the first publicly available set of plain images of surface defects in carbon look components.

Furthermore, the collected dataset contains real-world images, with realistic noise (all the images were acquired with a smartphone) and include all the defects relevant to the company that has been our case study. In this regard, our study focused on surface defects. Indeed, other applications might include different type of defects (including structural ones) and more noise. Therefore, to obtain more general results and detect different type of defects, the dataset should be expanded and a different training run. Moreover, the image resolution (224 × 224 pixels) and the collection of images in a real-world scenario, with a smartphone without standardizing the point of view, show the capability of our classifiers in low-quality conditions. However, to obtaion more general results about the robustness of the proposed models with even lower quality images, the dataset can be further expanded by adding artificial blur and noise[2] to the images.

Finally, the results suggest that the proposed classifiers could be added to the naked-eye quality control of carbon look components currently performed in the company that was our case study. However, before proceeding to production, a study dedicated to verifying computation times[3] and the image processing on handheld devices might be necessary in order to draw definitive conclusions in this regard. Furthermore, we obtained the presented results by testing different combinations of hyperparameters of the proposed models. Those based on DenseNet121 and Xception for the binary dataset, and DenseNet121 and VGG16 for the multi-class dataset, showed the best generalization capability. A more detailed investigation about the best hyperparameters specifically dedicated to such models might be worth conducting to find the optimal combination.

5. Conclusions

In this paper, we presented a comparison of ten end-to-end neural networks based on pre-trained CNNs, developed to classify surface defects in carbon look components (covered with CFRP) in a transfer learning fashion. We compared the proposed networks on a database of images from a real case study, with pictures of carbon look components for the automotive sector taken with a smartphone, under varying lighting conditions and from different points of view. In this way, we studied the use of the classification of plain images as a tool to automate or integrate a task currently performed by specialized operators with the naked eye. Our approach differs from other methodologies available in the literature as it does not use NDT techniques requiring dedicated settings, such as Infrared thermography.

The results are promising: the best model, the one based on DenseNet121, is capable of distinguishing between 900 images of components with recoverable defects, non-recoverable defects and those without any defect with a 97% accuracy. The other three models, those based on VGG16, VGG19, and Xception, exhibit a similar accuracy. Although the generality of such accuracy should be interpreted with caution due to the reasons explained in Section 4.3, the proposed end-to-end deep neural networks are adequate to classify surface defects in the carbon look components produced in "HP Composite Spa". The simplicity of the proposed methodology, based on end-to-end deep neural networks, and the classification accuracy with images taken from different points of view, with varying lighting conditions, make the quality control of carbon look components feasible. Given the possibility of porting CNNs into mobile devices [50,51], our approach can be integrated into handheld devices by deploying the classifier locally or in the cloud. Nevertheless, as highlighted in Section 4.3, a study dedicated to verifying computation times and the image

中文详解 / 原文第16页

局限与结论：能说到哪里就说到哪里（必须精读）

[1] 作者承认需要更多原始图片与不同部件验证。真实环境、不同照明是优点，但来源范围仍有限；模型可能对新工件、新设备或新材料表现下降。没有针对这些变化的测试，就不该声称已有跨域泛化。

[2] 分辨率224×224以及手持拍摄会影响细小缺陷。加入模糊、噪声可以用于鲁棒性实验，但这仍是模拟扰动；与新工厂真实采集是不同证据。

[3] 部署前还要检查计算时间、设备图像处理和工作流程。模型准确率只是系统的一部分；拍摄、筛选、复核、数据留档都会影响实际使用。

[4] 结论应写成：在作者这套汽车外观数据和评估设置中，预训练CNN分类表现较好。不能改写成"证明能识别飞机内部损伤"，也不能直接证明高铁与复合材料之间可以迁移。

[5] 你的下一步先把真实数据来源、标签和独立性弄清楚，再设计基线与对比。自检：从这篇论文你能借鉴什么，哪些结论还不能借用？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

processing on handheld devices might be necessary in order to draw definitive conclusions in this regard.

The source code[2] of the experiments, including the ten end-to-end networks, and the database of images collected for the tests are publicly available in dedicated open-access repositories. The database of images can be useful to benchmark other classification techniques. The source code of our experiments makes the research presented in this paper fully reproducible.

Author Contributions: Conceptualization, A.S. and V.C.; methodology, A.S., V.C. and P.S.; software, A.S. and P.S.; validation, S.T., N.F. and P.C.; data curation, A.S.; writing—original draft preparation, P.S.; writing—review and editing, S.T., N.F., P.C., A.B. and A.F.D.; supervision, A.B. and A.F.D. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable

Informed Consent Statement: Not applicable.

Data Availability Statement:[1] The dataset used in this study is publicly available at https://github.com/airtlab/surface-defect-classification-in-carbon-look-components-dataset (accessed on the 10th of July 2023). The source code of the experiments performed on such dataset is publicly available at https://github.com/airtlab/surface-defect-classification-in-carbon-look-components-using-deep-neural-networks
(accessed on the 10th of July 2023).

Acknowledgments: The authors thank HP Composites Spa for providing the data used in this research.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Lasi, H.; Fettke, P.; Kemper, H.G.; Feld, T.; Hoffmann, M. Industry 4.0. *Bus. Inf. Syst. Eng.* **2014**, *6*, 239–242. https://doi.org/10.1007/s12599-014-0334-4.
2. Lee, J.; Davari, H.; Singh, J.; Pandhare, V. Industrial Artificial Intelligence for industry 4.0-based manufacturing systems. *Manuf. Lett.* **2018**, *18*, 20–23. https://doi.org/10.1016/j.mfglet.2018.09.002.
3. Zenisek, J.; Holzinger, F.; Affenzeller, M. Machine learning based concept drift detection for predictive maintenance. *Comput. Ind. Eng.* **2019**, *137*, 106031. https://doi.org/10.1016/j.cie.2019.106031.
4. Zhou, X.; Hu, Y.; Liang, W.; Ma, J.; Jin, Q. Variational LSTM Enhanced Anomaly Detection for Industrial Big Data. *IEEE Trans. Ind. Inform.* **2021**, *17*, 3469–3477. https://doi.org/10.1109/TII.2020.3022432.
5. Gao, Z.; Dong, G.; Tang, Y.; Zhao, Y.F. Machine learning aided design of conformal cooling channels for injection molding. *J. Intell. Manuf.* **2021**, *34*, 1183–1201. https://doi.org/10.1007/s10845-021-01841-9.
6. Cadavid, J.P.U.; Lamouri, S.; Grabot, B.; Pellerin, R.; Fortin, A. Machine learning applied in production planning and control: A state-of-the-art in the era of industry 4.0. *J. Intell. Manuf.* **2020**, *31*, 1531–1558. https://doi.org/10.1007/s10845-019-01531-7.
7. Peres, R.S.; Barata, J.; Leitao, P.; Garcia, G. Multistage Quality Control Using Machine Learning in the Automotive Industry. *IEEE Access* **2019**, *7*, 79908–79916. https://doi.org/10.1109/ACCESS.2019.2923405.
8. Yang, J.; Li, S.; Wang, Z.; Dong, H.; Wang, J.; Tang, S. Using Deep Learning to Detect Defects in Manufacturing: A Comprehensive Survey and Current Challenges. *Materials* **2020**, *13*, 5755. https://doi.org/10.3390/ma13245755.
9. Obregon, J.; Hong, J.; Jung, J.Y. Rule-based explanations based on ensemble machine learning for detecting sink mark defects in the injection moulding process. *J. Manuf. Syst.* **2021**, *60*, 392–405. https://doi.org/10.1016/j.jmsy.2021.07.001.
10. Polenta, A.; Tomassini, S.; Falcionelli, N.; Contardo, P.; Dragoni, A.F.; Sernani, P. A Comparison of Machine Learning Techniques for the Quality Classification of Molded Products. *Information* **2022**, *13*, 272. https://doi.org/10.3390/info13060272.
11. Nash, W.; Drummond, T.; Birbilis, N. A review of deep learning in the study of materials degradation. *npj Mater. Degrad.* **2018**, *2*, 37. https://doi.org/10.1038/s41529-018-0058-x.
12. Jorge Aldave, I.; Venegas Bosom, P.; Vega González, L.; López de Santiago, I.; Vollheim, B.; Krausz, L.; Georges, M. Review of thermal imaging systems in composite defect detection. *Infrared Phys. Technol.* **2013**, *61*, 167–175. https://doi.org/10.1016/j.infrared.2013.07.009.
13. Poór, D.I.; Geier, N.; Pereszlai, C.; Xu, J. A critical review of the drilling of CFRP composites: Burr formation, characterisation and challenges. *Compos. Part B Eng.* **2021**, *223*, 109155. https://doi.org/10.1016/j.compositesb.2021.109155.
14. Amor, N.; Noman, M.T.; Petru, M. Classification of Textile Polymer Composites: Recent Trends and Challenges. *Polymers* **2021**, *13*, 2592. https://doi.org/10.3390/polym13162592.

中文详解 / 原文第17页

数据与代码声明：怎样复用研究材料（精读数据声明，略读作者分工）

[1] 本页给出数据仓库和实验代码仓库。数据仓库提供图片与类别，代码仓库提供作者的实验实现；它们不是同一件东西。下载数据不等于已经复现论文，跑通自己的ResNet18也不等于作者十模型十划分比较已经完成。

[2] 作者实验主要使用Keras/TensorFlow；你的程序使用PyTorch。移植时需要核对预处理、层定义、优化器默认参数、早停与划分，而不只是把模型名字换过来。

[3] 作者贡献、经费与利益声明本轮可以略读，但数据许可和使用条件需要保留。已公开材料也应注明来源，不能把作者原始图片当作自己采集的数据。

[4] 自检：如果缺少微调后权重，但提供完整训练代码，你为了复现还需要保存哪些设置和随机划分？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

15. Vedernikov, A.; Tucci, F.; Carlone, P.; Gusev, S.; Konev, S.; Firsov, D.; Akhatov, I.; Safonov, A. Effects of pulling speed on structural performance of L-shaped pultruded profiles. *Compos. Struct.* **2021**, *255*, 112967. https://doi.org/10.1016/j.compstruct.2020.112967.
16. Bang, H.T.; Park, S.; Jeon, H. Defect identification in composite materials via thermography and deep learning techniques. *Compos. Struct.* **2020**, *246*, 112405. https://doi.org/10.1016/j.compstruct.2020.112405.
17. Liu, K.; Li, Y.; Yang, J.; Liu, Y.; Yao, Y. Generative Principal Component Thermography for Enhanced Defect Detection and Analysis. *IEEE Trans. Instrum. Meas.* **2020**, *69*, 8261–8269. https://doi.org/10.1109/TIM.2020.2992873.
18. Fang, Q.; Ibarra-Castanedo, C.; Maldague, X. Automatic Defects Segmentation and Identification by Deep Learning Algorithm with Pulsed Thermography: Synthetic and Experimental Data. *Big Data Cogn. Comput.* **2021**, *5*, 9. https://doi.org/10.3390/bdcc5010009.
19. Wei, Z.; Fernandes, H.; Herrmann, H.G.; Tarpani, J.R.; Osman, A. A Deep Learning Method for the Impact Damage Segmentation of Curve-Shaped CFRP Specimens Inspected by Infrared Thermography. *Sensors* **2021**, *21*, 395. https://doi.org/10.3390/s21020395.
20. Pan, S.J.; Yang, Q. A Survey on Transfer Learning. *IEEE Trans. Knowl. Data Eng.* **2010**, *22*, 1345–1359. https://doi.org/10.1109/TKDE.2009.191.
21. Simonyan, K.; Zisserman, A. Very Deep Convolutional Networks for Large-Scale Image Recognition. *arXiv* **2015**, arXiv:1409.1556.
22. He, K.; Zhang, X.; Ren, S.; Sun, J. Identity Mappings in Deep Residual Networks. In Proceedings of the Computer Vision—ECCV 2016, Amsterdam, The Netherlands, 11–14 October 2016 ; Springer International Publishing: Cham, Switzerland, 2016; pp. 630–645. https://doi.org/10.1007/978-3-319-46493-0_38.
23. Szegedy, C.; Vanhoucke, V.; Ioffe, S.; Shlens, J.; Wojna, Z. Rethinking the Inception Architecture for Computer Vision. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 27–30 June 2016; pp. 2818–2826. https://doi.org/10.1109/CVPR.2016.308.
24. Sandler, M.; Howard, A.; Zhu, M.; Zhmoginov, A.; Chen, L.C. MobileNetV2: Inverted Residuals and Linear Bottlenecks. In Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–23 June 2018; pp. 4510–4520. https://doi.org/10.1109/CVPR.2018.00474.
25. Zoph, B.; Vasudevan, V.; Shlens, J.; Le, Q.V. Learning Transferable Architectures for Scalable Image Recognition. In Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018; pp. 8697–8710. https://doi.org/10.1109/CVPR.2018.00907.
26. Huang, G.; Liu, Z.; Van Der Maaten, L.; Weinberger, K.Q. Densely Connected Convolutional Networks. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017; pp. 2261–2269. https://doi.org/10.1109/CVPR.2017.243.
27. Chollet, F. Xception: Deep Learning with Depthwise Separable Convolutions. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017; pp. 1800–1807. https://doi.org/10.1109/CVPR.2017.195.
28. Bray, D.E.; McBride, D. Nondestructive Testing Techniques; NASA STI/Recon Technical Report A; Wiley: Hoboken, NJ, USA, 1992; Volume 93, p. 17573.
29. Gholizadeh, S. A review of non-destructive testing methods of composite materials. *Procedia Struct. Integr.* **2016**, *1*, 50–57. https://doi.org/10.1016/j.prostr.2016.02.008.
30. LeCun, Y.; Bengio, Y.; Hinton, G. Deep learning. *Nature* **2015**, *521*, 436–444. https://doi.org/10.1038/nature14539.
31. Liu, K.; Ma, Z.; Liu, Y.; Yang, J.; Yao, Y. Enhanced Defect Detection in Carbon Fiber Reinforced Polymer Composites via Generative Kernel Principal Component Thermography. *Polymers* **2021**, *13*, 825. https://doi.org/10.3390/polym13050825.
32. Ioffe, S.; Szegedy, C. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. *arXiv* **2015**, arXiv:abs/1502.03167.
33. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. In Proceedings of the Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015, Munich, Germany, 5–9 October 2015; Navab, N., Hornegger, J., Wells, W.M., Frangi, A.F., Eds.; Springer: Cham, Switzerland, 2015; pp. 234–241. https://doi.org/10.1007/978-3-319-24574-4_28.
34. Marani, R.; Palumbo, D.; Galietti, U.; D'Orazio, T. Deep learning for defect characterization in composite laminates inspected by step-heating thermography. *Opt. Lasers Eng.* **2021**, *145*, 106679. https://doi.org/10.1016/j.optlaseng.2021.106679.
35. Meng, M.; Chua, Y.J.; Wouterson, E.; Ong, C.P.K. Ultrasonic signal classification and imaging system for composite materials via deep convolutional neural networks. *Neurocomputing* **2017**, *257*, 128–135. https://doi.org/10.1016/j.neucom.2016.11.066.
36. Gong, Y.; Shao, H.; Luo, J.; Li, Z. A deep transfer learning model for inclusion defect detection of aeronautics composite materials. *Compos. Struct.* **2020**, *252*, 112681. https://doi.org/10.1016/j.compstruct.2020.112681.
37. Gao, Y.; Li, X.; Wang, X.V.; Wang, L.; Gao, L. A Review on Recent Advances in Vision-based Defect Recognition towards Industrial Intelligence. *J. Manuf. Syst.* **2022**, *62*, 753–766. https://doi.org/10.1016/j.jmsy.2021.05.008.
38. Yun, J.P.; Shin, W.C.; Koo, G.; Kim, M.S.; Lee, C.; Lee, S.J. Automated defect inspection system for metal surfaces based on deep learning and data augmentation. *J. Manuf. Syst.* **2020**, *55*, 317–324. https://doi.org/10.1016/j.jmsy.2020.03.009.
39. Chen, W.; Gao, Y.; Gao, L.; Li, X. A New Ensemble Approach based on Deep Convolutional Neural Networks for Steel Surface Defect classification. *Procedia CIRP* **2018**, *72*, 1069–1072. https://doi.org/10.1016/j.procir.2018.03.264.
40. Boikov, A.; Payor, V.; Savelev, R.; Kolesnikov, A. Synthetic Data Generation for Steel Defect Detection and Classification Using Deep Learning. *Symmetry* **2021**, *13*, 1176. https://doi.org/10.3390/sym13071176.

中文详解 / 原文第18页

参考文献：按问题反查，而不是全部读（略读）

[1] 参考文献是继续查证某个术语或方法的入口，不是你这一周必须读完的任务表。先标出与你实际数据最接近的一到两篇，例如同样普通RGB表面分类、或确实对应冲击热成像的数据论文。

[2] 引用别人的结论前应回到原论文检查数据和实验，不能把本篇相关工作的一句话当成完整证据。期刊分区、年份和学科口径另外核验；被本文引用不代表满足导师的二区要求。

[3] 本页不新增模型训练步骤。你可以用它检索不懂的具体概念，并在问题解决后返回主线。自检：下一篇文章会帮你解决哪个当前实际问题？

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。

41. Liu, J.; Wang, C.; Su, H.; Du, B.; Tao, D. Multistage GAN for Fabric Defect Detection. *IEEE Trans. Image Process.* **2020**, *29*, 3388–3400. https://doi.org/10.1109/TIP.2019.2959741.
42. Zhu, Z.; Han, G.; Jia, G.; Shu, L. Modified DenseNet for Automatic Fabric Defect Detection With Edge Computing for Minimizing Latency. *IEEE Internet Things J.* **2020**, *7*, 9623–9636. https://doi.org/10.1109/JIOT.2020.2983050.
43. Gao, Y.; Gao, L.; Li, X.; Wang, X.V. A Multilevel Information Fusion-Based Deep Learning Method for Vision-Based Defect Recognition. *IEEE Trans. Instrum. Meas.* **2020**, *69*, 3980–3991. https://doi.org/10.1109/TIM.2019.2947800.
44. Urbonas, A.; Raudonis, V.; Maskeliūnas, R.; Damaševičius, R. Automated Identification of Wood Veneer Surface Defects Using Faster Region-Based Convolutional Neural Network with Data Augmentation and Transfer Learning. *Appl. Sci.* **2019**, *9*, 4898. https://doi.org/10.3390/app9224898.
45. Shorten, C.; Khoshgoftaar, T.M. A survey on Image Data Augmentation for Deep Learning. *J. Big Data* **2019**, *6*, 60. https://doi.org/10.1186/s40537-019-0197-0.
46. Russakovsky, O.; Deng, J.; Su, H.; Krause, J.; Satheesh, S.; Ma, S.; Huang, Z.; Karpathy, A.; Khosla, A.; Bernstein, M.; et al. ImageNet Large Scale Visual Recognition Challenge. *Int. J. Comput. Vis.* **2015**, *115*, 211–252. https://doi.org/10.1007/s11263-015-0816-y.
47. Misra, I.; Zitnick, C.L.; Hebert, M. Shuffle and learn: Unsupervised learning using temporal order verification. In Proceedings of the Computer Vision—ECCV 2016, Amsterdam, The Netherlands, 11–14 October 2016; Springer: Cham, Switzerland, 2016; pp. 527–544. https://doi.org/10.1007/978-3-319-46448-0_32.
48. Weiss, K.; Khoshgoftaar, T.M.; Wang, D. A survey of transfer learning. *J. Big Data* **2016**, *3*, 9. https://doi.org/10.1186/s40537-016-0043-6.
49. Huh, M.; Agrawal, P.; Efros, A.A. What makes ImageNet good for transfer learning? *arXiv* **2016**, arXiv:1608.08614. .
50. Wu, J.; Leng, C.; Wang, Y.; Hu, Q.; Cheng, J. Quantized Convolutional Neural Networks for Mobile Devices. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 27–30 June 2016; pp. 4820–4828. https://doi.org/10.1109/CVPR.2016.521.
51. Li, D.; Wang, X.; Kong, D. DeepRebirth: Accelerating Deep Neural Network Execution on Mobile Devices. *Proc. AAAI Conf. Artif. Intell.* **2018**, *32*, 2322–2330. https://doi.org/10.1609/aaai.v32i1.11876.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

中文详解 / 原文第19页

参考文献与全文收束：形成一份周报（略读文献，完成回顾）

[1] 读完后用自己的语言串起问题、数据、标签、方法、评估、结果与局限，避免只记住一个97%的数字。至少亲自读第5—10页和第16页，再用图3、表3及一张主要结果图讲一次。

[2] 周报PPT应与这篇论文对应：不是最终研究论文答辩。模板确认后沿用统一风格；每张图写明来自作者还是自己的实验。你尚未复现的内容，要明确写“作者报告”。

[3] 自检清单：二分类negative到底包括什么？512特征怎样得到两个分数？训练验证测试各做什么？十次划分为什么需要？你的83.75%为什么不能直接与论文97%公平比较？如果这五个问题仍含糊，就先回到对应讲解页。

[4] 研究下一步仍保持开放：数据获取、相关论文、基本概念和基线实验服务于导师讨论；不在此刻把迁移课题定死。

黄色/红线：原文重点；红色编号：右栏解释。
中文为教学讲解，与作者原文区分。
原文 DOI: 10.3390/s23177607；CC BY 4.0。